UNIVERSITI TEKNOLOGI MALAYSIA

TASK 3

NETWORK COMMUNICATION

NO	NAME	MATRIC NO
1	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250
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3.1 Introduction to the needed devices

1. Switch

A switch is a networking device used to connect multiple devices, such as computers, printers, and servers, within a Local Area Network (LAN). It efficiently manages data traffic by directing information only to the intended device, improving network performance and reducing congestion

2. Patch panel

A patch panel is a hardware component that organizes and centralizes network cables from various locations in a building. It provides a single point where cables can be connected or disconnected, making it easier to manage, troubleshoot, and reconfigure network connections.

3. Modem

A modem is a device that enables internet connectivity by converting digital signals from a computer into analog signals suitable for transmission over telephone or cable lines, and vice versa. It serves as the interface between your local network and your internet service provider.

4. server

A server is a powerful computer or system that provides services, resources, or data to other devices on a network, known as clients. Servers can perform various functions, such as hosting websites, storing files, or managing applications, making them essential for business and organizational networks.

5. Cable

.A cable is a physical medium used to transmit data or power between devices in a network. Common examples include Ethernet cables, which are used to establish wired connections between devices and network hardware such as switches and routers.

6. Connector

A connector is the endpoint of a cable that allows it to interface with devices or other cables. For example, an RJ45 connector is commonly used for Ethernet cables in computer networks to ensure secure and stable connections.

7. Firewall

A firewall is a security system, either hardware or software, that monitors and controls incoming and outgoing network traffic. It acts as a barrier to block unauthorized access while permitting legitimate communication, protecting networks from cyber threats.

8. Router

A router is a networking device that connects different networks, such as a home network to the internet. It directs data packets between networks, ensuring that information is transmitted efficiently and reaches its intended destination.

9. Wireless access points

A wireless access point (WAP) is a device that allows wireless devices to connect to a wired network. It extends the reach of the network by providing Wi-Fi connectivity, enabling devices to access the network without the need for physical cables.

3.2 Table of devices

Devices	Description	Price per unit (RM)	Quantity	Total price (RM)
Switch 	**FS S3400-24T4SP** **High Availability and Multi-Service** : Support Support 4K Active VLAN, voice VLAN and Private VLAN **Error-Free Network Configuration :** Support Web management And support SNMP v1, v2c, v3. **Secure and Simplified Access for Users and IoT :** Support IEEE 802.1x, Radius and TACACS+ perfect security authentication mechanisms	2635	8	21080
Router 	**Cisco ISR4331-VSEC/K9 4331 Router, Gigabit Ethernet, VSEC License** **Performance**: Offers up to 100 Mbps of default performance and can scale up to 2 Gbps with performance-on-demand licenses. High throughput for WAN connections with integrated hardware	3733	3	11199

	acceleration. **Modularity**: Equipped with three built-in Gigabit Ethernet ports, two Network Interface Module (NIM) slots, and one Service Module (SM) slot. Allows customization with additional interfaces like fiber, Ethernet, or advanced services. **Advanced Security**: Includes Cisco IOS security features such as VPN, firewall, and advanced threat protection. Supports VLAN segmentation for secure traffic separation. **SD-WAN Ready**: Can integrate with Cisco SD-WAN for better application performance and simplified management of multiple network paths. **Built-In Redundancy**: Dual power supplies and modular connectivity ensure network reliability and flexibility.			

	Future-Ready: Supports virtualization and cloud integration, making it adaptable to evolving technological needs.			
Patch panel 	**FS 48-Port Cat6** **Toolless termination:** This Cat6 patch panel allows easy tool-free connections, reducing installation time and effort while providing high-quality structured cabling for academic networks. **Supports high-speed connections:** Optimized for gigabit Ethernet speeds, the 48 ports ensure seamless performance for data-intensive tasks. Ideal for academic environments that require reliable and high-speed networking **Cable management features:** Provides built-in cable management for a neat and organized installation, minimizing clutter and interference in network setups **Durability and scalability:** Made with high-quality materials, it supports easy scalability by accommodating additional network demands over time.	458.00	6	2748

Wireless access point (WAP) 	**Netgear WAX615 Cloud Managed Wireless Access Point - WiFi 6 Dual-Band AX3000** Wi-Fi 6 (802.11ax): Dual-band with a maximum throughput of 3 Gbps (600 Mbps on 2.4 GHz and 2.4 Gbps on 5 GHz). Supports high-density environments and simultaneous device connections. Cloud Management: Fully managed via Netgear Insight, providing remote access for monitoring, configuration, and updates. Ideal for IT administrators managing networks across multiple locations. High User Capacity: Supports up to 256 client devices per AP, making it suitable for labs, classrooms, and collaborative spaces. Security Features: WPA3 encryption, VLAN segmentation, and enhanced client isolation for secure connections. Suitable for areas handling sensitive data,	1,049	10	10490

	such as administrative offices and research labs. PoE+ Support: Power over Ethernet (802.3at) enables easy deployment without additional power cables.			
Modem 	**ARRIS SURFboard Gigabit DOCSIS** 3.1 Cable Modem **DOCSIS 3.1 cable modem :** backward compatible with DOCSIS 3.0. Supports IPv4 and IPv6 Internet browsing standards. - As a Docsis 3.1 32X8 Modem capable of up to 10 Gbps download speeds. As a Docsis 3.0 32x8 Modem is capable of up to 1.4 Gbps download speeds. **Gigabit Ethernet ports :** Capable of supporting 2 Home Networks, both capable of gigabit speeds.	1007.23	2	2014.46
Server 	**Dell PowerEdge R750 Rack Server** Processor: Intel® Xeon® Silver 4310 2.1G, 12C/24T, 10.4GT/s, 18M Cache, Turbo, HT (120W) DDR4-2666 RAM : 64GB RDIMM, 3200MT/s, Dual Rank, 16Gb Memory: 480GB SSD SATA Read Intensive 6Gbps 512	29,818.77	8	238550.00

	2.5in Hot-plug AG Drive, 1 DWPD ● Supports 8 channels per CPU, up to 32 DDR4 DIMMs at 3200 MT/s DIMM speed ● Ideal for traditional corporate IT, database and analytics, VDI, and AI/ML and Inferencing			
Ethernet Cable 	**50106ASFR - BELDEN Industrial Ethernet CAT6A cable, S/FTP, LSZH, Anti Rodent, Anti Termite, 4 Pair, AWG 23,Indoor** ● Conductor Material : Bare Copper ● Insulation material : Polyethylene ● Cabling : 4 pairs twisted together with central spline + overlapping polyester foil over cable core ● Max. Capacitance Unbalance : 160 pF/100m ● Max. Delay : 538 ns/100m ● Voltage Rating [V] : Max. 72 V DC ● Environmental Space : Indoor	6.71/m	10	671.00

3.3 reflection

3.3.1 Are you surprised by the prices? How were you surprised?

We found ourselves a bit taken aback by the overall pricing of the networking devices needed for our building. While we anticipated the cost of the server and had budgeted for it accordingly, the price of the firewall caught us off guard. This unexpected expense has caused us to reassess our budget and look for alternative solutions to meet our requirements without compromising on security.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

Yes. We have always considered cost as a factor for choosing networking devices. Fortunately, our budget is pretty sufficient for us to get all of the devices that we need. We get to choose the best devices that do not exceed our budget. Although we have quite a big budget, that does not mean we can go overboard when choosing the devices. We did a thorough research in order to not spend our budget on devices that are expensive but not suitable for the building as this will result in more problems in the future that might cost us more money.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Switch:

For the switches used in this project, we have chosen FS S3400-24T4FP. The major difference between FS S3400-24T4FP with other different brands we can compare is TP- Link TL - SG1024.

Feature	FS S3400-24T4SP	TP- Link TL - SG1024
Packet Buffer Size	500 KB	512 kb
Cooling	3 Built-in Fans	Fanless
Traffic Handling	High	Limited
Power Consumption	400 W	13.8 W
Power Efficiency	High	Low
Price (RM)	Rm 2635	

The FS S3400-24T4SP is significantly more expensive than the TP-Link TL-SG1024, but it offers superior features necessary for an academic institution. Its robust cooling system, including three built-in fans, ensures stability during heavy network loads, which is crucial for maintaining reliable performance. Additionally, the FS S3400-24T4SP handles high traffic efficiently due to its advanced technology and larger packet buffer size.

On the other hand, while the TP-Link TL-SG1024 is more affordable and has a slightly larger packet buffer (512 KB), its lack of cooling fans and lower traffic handling capabilities make it less suitable for a high-demand academic environment. Despite the higher cost, the FS S3400-24T4SP is a better long-term investment for institutions requiring reliable, high-performance networking.

Router:

We have chosen to install the Cisco ISR4331-VSEC/K9 4331 Router in the building. Another router from a different brand that we can compare to is the Juniper SRX345-DC SRX345 Router.

Feature	Cisco ISR4331-VSEC/K9 4331	Juniper SRX345-DC SRX345
Performance	100 Mbps to 300 Mbps (with services)	Up to 5 Gbps (basic forwarding), 800 Mbps with services
Security Features	Advanced (VSEC license for VPN, firewall)	Enterprise-grade (UTM, IPS, VPN, firewall)
Scalability	Modular design for additional services	Non-modular but supports clustering
Interfaces	3 Gigabit Ethernet ports (expandable via NIM)	16 Gigabit Ethernet ports
Target Audience	Enterprise, educational, and branch networks	Enterprise and medium-sized branch networks
Redundancy	Supported via external modules	Built-in clustering for high availability
Security Add-Ons	VPN, Firewall, IPSec with VSEC license	Unified Threat Management (UTM), Anti-virus

We chose the Cisco ISR4331-VSEC/K9 because it's built to handle a wide range of needs while being flexible for the future unlike the Juniper SRX345, which is built with the focus of excellency for security and speed. Although speed and security are key factors in choosing networking devices, we still proceeded with Cisco ISR4331 because it is still provided with top-notch security features and performance and it comes with a much lower price than Juniper.

Patch panel :

For the patch panel used in this project, we have chosen FS 48-Port Cat6 Patch Panel. The major difference between FS 48-Port Cat6 Patch Panel with other different brands we can compare is Panduit Patch Panel.

Feature	**FS 48-Port Cat6 Patch Panel**	**Panduit Patch Panel**
Connector type	RJ45	RJ45
Cable type	Ethernet (Cat6, Cat5e, Cat7)	Ethernet (Cat6)
Cable management	Detachable back bar with zip ties for organized cable handling.	Basic cable management without advanced features.
10G support	Yes	No
Dimension	20 x 4.5 x 2.5 inches	19 x 1.77 x 0.4 inches
Price	RM 458.00	

The FS 48-Port Cat6 Patch Panel offers advanced features that make it a superior choice for a high-performance academic network. With support for multiple cable types, 10G bandwidth, and advanced cable management options like detachable back bars and zip ties, it ensures durability and ease of use during installation and maintenance. In comparison, the Panduit Patch Panel is a more basic option, offering standard RJ45 connections and Cat6 cable support but lacking advanced cable management and 10G compatibility. Although the Panduit Patch Panel is more affordable at RM 450.00, its limited features and scalability make it less suitable for long-term or high-demand setups. The FS 48-Port Cat6 Patch Panel, despite its higher price of RM 879.00, is a more reliable and future-proof option for robust academic infrastructure.

Wireless Access Point:

We have chosen Netgear WAX615 WAP for our building, the other brand that we chose to compare their major differences is TP-Link Omada EAP670 WAP.

Feature	Netgear WAX615	TP-Link Omada EAP670
WiFi Standard	Wi-Fi 6 (802.11ax)	Wi-Fi 6 (802.11ax)
Max Throughput	3.0 Gbps (600 Mbps on 2.4 GHz, 2.4 Gbps on 5 GHz)	5.4 Gbps (574 Mbps on 2.4 GHz, 4.8 Gbps on 5 GHz)
Client Capacity	Up to 256 clients per AP	Up to 512 clients per AP
Cloud Management	Netgear Insight	TP-Link Omada SDN
Poe Support	PoE+ (802.3at)	PoE+ (802.3at)
Coverage Area	~30 meters indoors	~30–40 meters indoors
Security Features	WPA3, VLAN segmentation, Client Isolation	WPA3, VLAN segmentation, Client Isolation
Warranty	Limited Lifetime Warranty	2-3 years standard warranty

After comparing the Netgear WAX615 and the TP-Link Omada EAP670, We have decided that the Netgear WAX615 is the better choice for the building. While TP-Link offers a slightly lower price and higher theoretical throughput, the Netgear WAX615 stands out in areas that matter most to us: reliability, long-term support, and robust security features.

The Netgear WAX615 provides a lifetime warranty, ensuring peace of mind and a better return on investment. Its performance in high-density environments, like labs and classrooms, is crucial for ensuring seamless connectivity during peak usage.

Although it costs a bit more, the additional features and reliability make it worth the investment. Netgear WAX615 is the ideal solution for building a network that is secure, scalable, and capable of supporting the demands of modern education spaces.

Modem:

We selected ARRIS SURFboard Gigabit DOCSIS 3.1 Cable Modem for the modem in this project. The major difference between ARRIS SURFboard Gigabit DOCSIS 3.1 Cable Modem and other different brands we can compare is Motorola MB8600 DOCSIS 3.1 Cable Modem

Specification	ARRIS SURFboard SB8200	Motorola MB8600
DOCSIS Standard	DOCSIS 3.1, also compatible with DOCSIS 3.0	DOCSIS 3.1, DOCSIS 3.0 32×8
Top Internet Speed Supported	Up to 2 Gbps*	Up to 6 Gbps
Ethernet Ports	Two 1-Gigabit Ethernet ports	1x 1Gbps LAN
Compatibility with Providers	Compatible with major U.S. cable providers	Comcast Xfinity, Cox, and others
Additional Features	Ideal for 4K Ultra-HD video streaming and VR gaming, Energy-Efficient Ethernet	WAN Link Aggregation, Multi-Static-IP support

Based on the table above, the ARRIS SURFboard SB8200 is a preferred choice due to its dual 1-Gigabit Ethernet ports, offering the flexibility of setting up two networks or a failover connection. It is highly compatible with major U.S. cable providers, ensuring broad usability. The modem excels in handling high-demand applications like 4K Ultra-HD video streaming and VR gaming, catering to users with intensive internet usage. Additionally, its energy-efficient ethernet feature presents potential cost savings. The inclusion of an app for easy modem setup further enhances its user-friendliness, making it an attractive option for those seeking a blend of performance, versatility, and convenience.

Server :

For the server that will be used in this project, we choose the Dell PowerEdge R750 Rack Server. The difference between Dell PowerEdge R750 Rack Server with other different brands we can compare is Cisco UCS C220 M6 Rack Server.

Specification	Dell PowerEdge R750 Rack Server	Cisco UCS C220 M6 Rack Server
Design and form actor	2U rack server	1U rack server
Processor	3rd Generation Intel Xeon Scalable processors	3rd Generation Intel Xeon Scalable processors
Memory	32 DIMMs for a maximum of 8TB of DDR4 or DDR5 memory.	32 DIMMs for a maximum of 8TB of DDR4 memory.
Storage	up to 24 drive bays	10 SFF drives
Expansion Slots	Up to 8 PCIe Gen4 slots	Up to 3 PCIe Gen4
Price (RM)	RM 29,818.77	RM 29,567.23

As we can see, the Dell PowerEdge R750 Rack Server is more expensive than Cisco UCS C220 M6 Rack Server, but it has more benefits for expandability, storage options, and versatility for intensive workloads. As we can see, the Dell PowerEdge R750 Rack Server expansion slots provide excellent expandability with Up to 8 PCIe Gen4 slots. Even though we buy two Cisco UCS C220 M6 Rack Server to get a 2U rack server, it still couldn’t beat the Dell PowerEdge R750 Rack Server expandability. Other than that, the storage is the Dell PowerEdge R750 Rack Server up to 16 x 2.5" drives or 8 x 3.5" drives (SAS/SATA/NVMe). It provides more flexibility for scalable storage solutions. Meanwhile Cisco UCS C220 M6 Rack Server supports up to 10 x 2.5" drives or 4 x NVMe drives. It has fewer storage options than the PowerEdge, focusing more on high-speed NVMe setups.

Fiber Optic Cable:

For the Fiber optic cable , we choose the RiteAV - 1M OM4 - 40Gb Multimode (50/125) - Duplex - Fiber Optic Cable - LC to LC. The difference between RiteAV - 1M OM4 - 40Gb Multimode (50/125) - Duplex - Fiber Optic Cable - LC to LC with other different brands we can compare is 7 Meter 10Gb OM3 Multimode Duplex Fiber Optic Cable (50/125) - LC to LC.

Specification	OM4 Fiber Optic Cable	OM3 Fiber Optic Cable
Bandwidth	Supports 10 Gbps up to 550 meters.	Supports 10 Gbps up to 300 meters.
Data Rates	10 Gbps and higher data rates	10 Gbps and higher data rates
Transmission Distances	suitable for high-speed networks, especially in data centers.	suitable for high-speed networks, especially in data centers.
Core size	50 microns	50 microns
Price	54/m	81/7m

Compared to previous multimode fibers like OM3, OM4 (Optical Multimode 4) is a form of multimode fiber optic cable that is intended to provide faster data speeds and longer distances. OM4 are optimized for 40GbE and 100GbE applications and support faster data transfer rates due to their higher bandwidth capabilities. Their backward compatibility enables smooth upgrades in current systems, and their increased reach offers flexibility in network architecture. For short-distance applications, OM4 cables are also affordable, balancing economy and performance. In data centers and high-performance environments, where reliable communication and fast data rates are more important than ever, the importance of OM4 connectors is further highlighted. In a world where technology is developing at an unstoppable rate, OM4 patch cables not only meet but also surpass the demands of networking enthusiasts and professionals.

Connector:

For connector we choose Register Jack 45 or RJ45 connector. The difference between RJ45 and other connectors we can compare is LC, SC, ST Fiber Optic Connectors.

Features	Register Jack 45	LC, SC, ST Fiber Optic Connectors
Primary use	Networking	High-speed networking
Speed	Up to 10Gbps	Up to 100 Gbps
Range	Moderate	Long
Cost	Low	High
Ease of Use	Easy	Specialized

By comparing these two connectors, both Register Jack 45 and LC, SC, ST Fiber Optic Connectors are for networking but LC, SC, ST Fiber Optic Connectors are high-speed networking for primary use. It is because speed for LC, SC, ST Fiber Optic Connectors is up to 100Gb per seconds meanwhile Register Jack 45 is up to 10Gb per second. However, despite its high speed, Register Jack 45 is still the choice because the range is moderate. It is suitable for the student area. Also, LC, SC, ST Fiber Optic Connectors required high cost while Register Jack 45 needed low cost. Plus, it is easy for students to use Register Jack 45 while LC, SC, ST Fiber Optic Connectors need specialized use.

Firewall

For firewalls, we choose Fortinet FortiGate. The difference between Fortinet FortiGate and other firewalls we can compare is Cisco ASA.

Features	Fortinet FortiGate	Cisco ASA
Security type	Next-Generation Firewall (NGFW)	Traditional Firewall (with NGFW upgrades via Firepower)
Performance	Optimized with custom FortiASIC processors	Reliable but may require upgrades for NGFW performance
Ease of Management	Unified Fortinet Security Fabric for seamless integration	Integrates well with Cisco ecosystem but may require multiple tools
Cost	Typically more cost-effective	Higher upfront costs, especially with NGFW upgrades
Cloud Compatibility	Extensive (AWS, Azure, hybrid environments)	Supported but better with Firepower
VPN Features	Strong site-to-site and remote VPN support	Industry-standard VPN features (AnyConnect)

As we can see from the table Fortinet FortiGate offers Next-Generation Firewall (NGFW) features by default, such as intrusion prevention, application control while Cisco ASA is primarily a traditional firewall with stateful inspection. To be said, Fortinet FortiGate is better for organizations looking for an all-in-one solution with modern security features, cost-effectiveness, and excellent performance in high-demand environments. Cisco ASA is ideal for businesses with existing Cisco ecosystems or those prioritizing robust traditional firewalls and VPNs.

MEETING MINUTE 1

DATE/TIME	15th november 2024	
LOCATION	ONLINE - GMEET	
AGENDA	1) Discuss the question given for Task 3 2) List down devices that is needed 3) Divide task for each member	
Meeting MC	YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE		
NAME	**TIME**	**REASON FOR ABSENCE**
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	2:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM	2:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN	2:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR	2:00 PM	-

MINUTES

NO.	**ITEM DISCUSSED**	**IDEAS/SUGGESTIONS AND PERSON GIVING IT**	**PERSON IN CHARGE & DATE**
1	Discuss the materials given for Task 3	i) Syarifah Dania suggested all of the team members go through the questions and determine the requirements needed to complete Task3 .	All members (15/11)

		ii) All members read the materials given and identify the requirements	
2	Provide devices	i) Yasmin Batrisyia and Syarifah Dania suggest each member to search for devices to achieve the objective of fulfilling the requirements and needs of the organization.. ii) All members find the suitable devices.	Yasmin Batrisyia (15/11) Muhammad Mukhritz Al Iman (15/11)
3	Next meeting	i) All members agreed to conduct another meeting to discuss more about our research. ii) Agenda of the next meeting was decided. iii) Syarifah Dania reminded all members to do some research about the project to ensure that everyone will have more understanding before next meeting	All members (15/11)
5	Meeting ended	After completing the dividing task between each member, Ahmad Adib ended the meeting.	Ahmad Adib (15/11)

MEETING MINUTE 2

DATE/TIME		18nd of November 2024	
LOCATION		Ar-Rayyan Cafe, KTDI	
AGENDA		1) Analyse device chosen from each member 2) Answer the question 3) Finalise the report.	
Meeting MC		YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE			
NAME		**TIME**	**REASON FOR ABSENCE**
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		9:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		9:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		9:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		9:00 PM	-
MINUTES			
NO.	**ITEM DISCUSSED**	**IDEAS/SUGGESTIONS AND PERSON GIVING IT**	**PERSON IN CHARGE & DATE**
1	Analyse question prepared by each member	i) Muhammad Mukhritz prepared list of all devices that has been prepared by each member ii) All members discuss the suitability of the device choosen	Muhammad Mukhritz Al Iman (18/11)

2	Task Distribution for each members	i) Muhammad Mukhritz prepared a poll for all members to choose which part that they want to do. ii) Each member chose do the part given	Syarifah Dania (18/11) Muhammad Mukhritz Al Iman (18/11)
3	Reporting	i) Task distribution for the report was done ii) Cover page, table of content and Introduction were done by Muhammad Mukhritz. iii) Meeting minutes was done by Syarifah Dania.	All members (18/11)
6	Meeting ended	Yasmin Batrisyia ended the meeting once the report was completed.	Yasmin Batrisyia (18/11)

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